

SQL JOINS PRACTICE WORKBOOK

Questions with SQL Queries and Output

Database Creation and Table Setup

Create Tables

```
CREATE TABLE departments (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50),  
    location VARCHAR(50)  
);
```

```
CREATE TABLE employees (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    dept_id INT,  
    salary INT,  
    city VARCHAR(50)  
);
```

```
CREATE TABLE projects (  
    proj_id INT PRIMARY KEY,  
    proj_name VARCHAR(50),  
    dept_id INT,  
    budget INT  
);
```

Insert Data

INSERT INTO departments VALUES

**(10,'HR','Jaipur'),
(20,'IT','Bangalore'),
(30,'Finance','Mumbai'),
(40,'Marketing','Delhi'),
(50,'Legal','Pune');**

INSERT INTO employees VALUES

**(101,'Amit',10,45000,'Jaipur'),
(102,'Bhavna',20,72000,'Bangalore'),
(103,'Chirag',20,68000,'Bangalore'),
(104,'Divya',30,55000,'Mumbai'),
(105,'Esha',NULL,39000,'Delhi'),
(106,'Farhan',40,61000,'Delhi'),
(107,'Gaurav',60,50000,'Kolkata'),
(108,'Hina',30,83000,'Mumbai');**

INSERT INTO projects VALUES

**(1,'Payroll Revamp',10,120000),
(2,'Cloud Migration',20,500000),
(3,'Mobile App',20,300000),
(4,'Audit Automation',30,250000),
(5,'Brand Refresh',40,180000),
(6,'Data Lake',70,400000);**

Question No. 1

List each employee with their department name and location.

```
47
48  -- 1
49
50 • SELECT e.emp_id, e.emp_name, d.dept_name, d.location
51 FROM employees e
52 INNER JOIN departments d
53 ON e.dept_id = d.dept_id;
54
```

Result Grid

	emp_id	emp_name	dept_name	location
▶	101	Amit	HR	Jaipur
	102	Bhavna	IT	Bangalore
	103	Chirag	IT	Bangalore
	104	Divya	Finance	Mumbai
	106	Farhan	Marketing	Delhi
	108	Hina	Finance	Mumbai

Question No. 2

Show every project along with the name of the department that owns it.

```

55  -- 2
56
57 • SELECT p.proj_id, p.proj_name, d.dept_name, p.budget
58 FROM projects p
59 INNER JOIN departments d
60 ON p.dept_id = d.dept_id;
61

```

Result Grid				
		Filter Rows:	Export:	Wrap Cell Content:
	proj_id	proj_name	dept_name	budget
▶	1	Payroll Revamp	HR	120000
	2	Cloud Migration	IT	500000
	3	Mobile App	IT	300000
	4	Audit Automation	Finance	250000
	5	Brand Refresh	Marketing	180000

Question No. 3

Show employee names together with the projects run by their own department.

```

62  -- 3
63
64 • SELECT e.emp_name, d.dept_name, p.proj_name
65 FROM employees e
66 INNER JOIN departments d
67 ON e.dept_id = d.dept_id
68 INNER JOIN projects p
69 ON d.dept_id = p.dept_id;

```

Result Grid			
		Filter Rows:	Export:
		Wrap Cell Content:	
	emp_name	dept_name	proj_name
▶	Amit	HR	Payroll Revamp
	Bhavna	IT	Mobile App
	Bhavna	IT	Cloud Migration
	Chirag	IT	Mobile App
	Chirag	IT	Cloud Migration
	Divya	Finance	Audit Automation
	Farhan	Marketing	Brand Refresh
	Hina	Finance	Audit Automation

Question No. 4

List ALL employees with their department name; show NULL when the employee has no matching department.

```

71  -- 4
72
73 • SELECT e.emp_id, e.emp_name, e.dept_id, d.dept_name
74 FROM employees e
75 LEFT JOIN departments d
76 ON e.dept_id = d.dept_id;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
emp_id	emp_name	dept_id	dept_name
101	Amit	10	HR
102	Bhavna	20	IT
103	Chirag	20	IT
104	Divya	30	Finance
105	Esha	NULL	NULL
106	Farhan	40	Marketing
107	Gaurav	60	NULL
108	Hina	30	Finance

Question No. 5

List ALL departments with their projects; departments with no project must still appear.

```
78 -- 5
79
80 • SELECT d.dept_id, d.dept_name, p.proj_name, p.budget
81 FROM departments d
82 LEFT JOIN projects p
83 ON d.dept_id = p.dept_id;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
dept_id	dept_name	proj_name	budget
10	HR	Payroll Revamp	120000
20	IT	Mobile App	300000
20	IT	Cloud Migration	500000
30	Finance	Audit Automation	250000
40	Marketing	Brand Refresh	180000
50	Legal	NULL	NULL

Question No. 6

Find only those employees who do NOT belong to any valid department.

```
85 -- 6
86
87 • SELECT e.emp_id, e.emp_name, e.dept_id
88 FROM employees e
89 LEFT JOIN departments d
90 ON e.dept_id = d.dept_id
91 WHERE d.dept_id IS NULL;
92
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
emp_id	emp_name	dept_id	
105	Esha	NULL	
107	Gaurav	60	

Question No. 7

List ALL departments and any employees in them, using a RIGHT JOIN with employees on the left.

```
--
93  -- 7
94
95 • SELECT e.emp_name, d.dept_id, d.dept_name
96     FROM employees e
97     RIGHT JOIN departments d
98     ON e.dept_id = d.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	emp_name	dept_id	dept_name
▶	Amit	10	HR
	Chirag	20	IT
	Bhavna	20	IT
	Hina	30	Finance
	Divya	30	Finance
	Farhan	40	Marketing
	NULL	50	Legal

Question No. 8

List ALL projects and the department that owns them, keeping projects that point to a missing department.

```
100  -- 8
101
102 • SELECT d.dept_name, p.proj_id, p.proj_name, p.dept_id
103     FROM departments d
104     RIGHT JOIN projects p
105     ON d.dept_id = p.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	dept_name	proj_id	proj_name	dept_id
▶	HR	1	Payroll Revamp	10
	IT	2	Cloud Migration	20
	IT	3	Mobile App	20
	Finance	4	Audit Automation	30
	Marketing	5	Brand Refresh	40
	NULL	6	Data Lake	70

Question No. 9

Find departments that currently have NO employee assigned.

```
107 -- 9
108
109 • SELECT d.dept_id, d.dept_name, d.location
110 FROM departments d
111 LEFT JOIN employees e
112 ON d.dept_id = e.dept_id
113 WHERE e.emp_id IS NULL;
114
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	dept_id	dept_name	location
▶	50	Legal	Pune

Question No. 10

Produce every possible pairing of the Finance/Legal departments with employees earning above 70000.

```
115 -- 10
116
117 • SELECT e.emp_name, e.salary, d.dept_name
118 FROM employees e
119 CROSS JOIN departments d
120 WHERE e.salary > 70000
121 AND d.dept_name IN ('Finance', 'Legal');
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	emp_name	salary	dept_name
▶	Bhavna	72000	Legal
	Bhavna	72000	Finance
	Hina	83000	Legal
	Hina	83000	Finance

Question No. 11

Pair every project having a budget of at least 400000 with every department located in Mumbai or Pune.

```
123  -- 11
124
125 • SELECT p.proj_name, p.budget, d.dept_name, d.location
126 FROM projects p
127 CROSS JOIN departments d
128 WHERE p.budget >= 400000
129 AND d.location IN ('Mumbai', 'Pune');
130
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	proj_name	budget	dept_name	location
▶	Data Lake	400000	Finance	Mumbai
	Cloud Migration	500000	Finance	Mumbai
	Data Lake	400000	Legal	Pune
	Cloud Migration	500000	Legal	Pune

Question No. 12

How many total row combinations result from CROSS JOIN of employees and departments? Show the count.

```
131  -- 12
132
133 • SELECT COUNT(*) AS total_combinations
134 FROM employees
135 CROSS JOIN departments;
136
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	total_combinations
▶	40

Question No. 13

Show employees whose salary is greater than 60000.

```
137  -- 13
138
139 • SELECT *
140 FROM employees
141 WHERE salary > 60000;
142
```

Result Grid

emp_id	emp_name	dept_id	salary	city
102	Bhavna	20	72000	Bangalore
103	Chirag	20	68000	Bangalore
106	Farhan	40	61000	Delhi
108	Hina	30	83000	Mumbai
NULL	NULL	NULL	NULL	NULL

Question No. 14

Show all employees based in Delhi or Mumbai.

```
143  -- 14
144
145 • SELECT emp_id, emp_name, salary, city
146 FROM employees
147 WHERE city IN ('Delhi', 'Mumbai');
148
```

Result Grid

emp_id	emp_name	salary	city
104	Divya	55000	Mumbai
105	Esha	39000	Delhi
106	Farhan	61000	Delhi
108	Hina	83000	Mumbai
NULL	NULL	NULL	NULL

Question No. 15

Show employees whose dept_id is NULL (unassigned employees).

```
140
149  -- 15
150
151 • SELECT emp_id, emp_name, dept_id, salary
152 FROM employees
153 WHERE dept_id IS NULL;
154
```

Result Grid

	emp_id	emp_name	dept_id	salary
▶	105	Esha	NULL	39000
*	NULL	NULL	NULL	NULL

Question No. 16

Show departments located in Bangalore or Delhi.

```
155  -- 16
156
157 • SELECT *
158 FROM departments
159 WHERE location IN ('Bangalore', 'Delhi');
160
```

Result Grid

	dept_id	dept_name	location
▶	20	IT	Bangalore
	40	Marketing	Delhi
*	NULL	NULL	NULL

Question No. 17

Show departments whose dept_id is 30 or higher.

```
161  -- 17
```

```
162
```

```
163 • SELECT *
```

```
164 FROM departments
```

```
165 WHERE dept_id >= 30;
```

```
166
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
dept_id	dept_name	location		
30	Finance	Mumbai		
40	Marketing	Delhi		
50	Legal	Pune		
NULL	NULL	NULL		

Question No. 18

Show departments whose name starts with the letter "F" or "L".

```
167  -- 18
```

```
168
```

```
169 • SELECT *
```

```
170 FROM departments
```

```
171 WHERE dept_name LIKE 'F%'
```

```
172 OR dept_name LIKE 'L%';
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
dept_id	dept_name	location		
30	Finance	Mumbai		
50	Legal	Pune		
NULL	NULL	NULL		

Question No. 19

Show projects with a budget between 200000 and 400000 (inclusive).

```
174  -- 19
175
176 • SELECT *
177 FROM projects
178 WHERE budget BETWEEN 200000 AND 400000;
```

Result Grid

proj_id	proj_name	dept_id	budget
3	Mobile App	20	300000
4	Audit Automation	30	250000
6	Data Lake	70	400000
*	NULL	NULL	NULL

Question No. 20

Show all projects that belong to dept_id 20.

```
180  -- 20
181
182 • SELECT *
183 FROM projects
184 WHERE dept_id = 20;
```

Result Grid

proj_id	proj_name	dept_id	budget
2	Cloud Migration	20	500000
3	Mobile App	20	300000
*	NULL	NULL	NULL

Question No. 21

Show projects whose name contains the word "a" and budget is under 300000.

```

186 -- 21
187
188 • SELECT proj_id, proj_name, budget
189 FROM projects
190 WHERE proj_name LIKE '%a%'
191 AND budget < 300000;

```

Result Grid			
Filter Rows:			
Edit: Export/Import: Wrap Cell Content:			
proj_id	proj_name	budget	
1	Payroll Revamp	120000	
4	Audit Automation	250000	
5	Brand Refresh	180000	
* NULL	NULL	NULL	

Question No. 22

Show employee name, department name, location and project name for all matching rows across all three tables.

```

193 -- 22
194
195 • SELECT e.emp_name, d.dept_name, d.location, p.proj_name, p.budget
196 FROM employees e
197 INNER JOIN departments d
198 ON e.dept_id = d.dept_id
199 INNER JOIN projects p
200 ON d.dept_id = p.dept_id;

```

Result Grid					
Filter Rows:					
Export: Wrap Cell Content:					
emp_name	dept_name	location	proj_name	budget	
Amit	HR	Jaipur	Payroll Revamp	120000	
Bhavna	IT	Bangalore	Mobile App	300000	
Bhavna	IT	Bangalore	Cloud Migration	500000	
Chirag	IT	Bangalore	Mobile App	300000	
Chirag	IT	Bangalore	Cloud Migration	500000	
Divya	Finance	Mumbai	Audit Automation	250000	
Farhan	Marketing	Delhi	Brand Refresh	180000	
Hina	Finance	Mumbai	Audit Automation	250000	

Question No. 23

Show ALL employees, plus department and project details where available (keep employees even with no dept/project).

```
202 -- 23
203
204 • SELECT e.emp_id, e.emp_name, d.dept_name, p.proj_name
205 FROM employees e
206 LEFT JOIN departments d
207 ON e.dept_id = d.dept_id
208 LEFT JOIN projects p
209 ON d.dept_id = p.dept_id;
```

emp_id	emp_name	dept_name	proj_name
101	Amit	HR	Payroll Revamp
102	Bhavna	IT	Mobile App
102	Bhavna	IT	Cloud Migration
103	Chirag	IT	Mobile App
103	Chirag	IT	Cloud Migration
104	Divya	Finance	Audit Automation
105	Esha	NULL	NULL
106	Farhan	Marketing	Brand Refresh
107	Gaurav	NULL	NULL
108	Hina	Finance	Audit Automation

Question No. 24

Show employees earning more than 60000 along with department and any project over 250000 budget.

```
211 -- 24
212
213 • SELECT e.emp_name, e.salary, d.dept_name, p.proj_name, p.budget
214 FROM employees e
215 INNER JOIN departments d
216 ON e.dept_id = d.dept_id
217 INNER JOIN projects p
218 ON d.dept_id = p.dept_id
219 WHERE e.salary > 60000
220 AND p.budget > 250000;
```

emp_name	salary	dept_name	proj_name	budget
Bhavna	72000	IT	Mobile App	300000
Bhavna	72000	IT	Cloud Migration	500000
Chirag	68000	IT	Mobile App	300000
Chirag	68000	IT	Cloud Migration	500000

Question No. 25

List ALL departments with their employees and projects, including departments having neither.

```
222  -- 25
223
224  • SELECT d.dept_id, d.dept_name, e.emp_name, p.proj_name
225  FROM departments d
226  LEFT JOIN employees e
227  ON d.dept_id = e.dept_id
228  LEFT JOIN projects p
229  ON d.dept_id = p.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	dept_id	dept_name	emp_name	proj_name
▶	10	HR	Amit	Payroll Revamp
	20	IT	Chirag	Mobile App
	20	IT	Chirag	Cloud Migration
	20	IT	Bhavna	Mobile App
	20	IT	Bhavna	Cloud Migration
	30	Finance	Hina	Audit Automation
	30	Finance	Divya	Audit Automation
	40	Marketing	Farhan	Brand Refresh
	50	Legal	NULL	NULL

Question No. 26

Show employees who work in a department located in Bangalore or Mumbai, along with the projects of that department.

```
231  -- 26
232
233  • SELECT e.emp_name, d.location, p.proj_name
234  FROM employees e
235  INNER JOIN departments d
236  ON e.dept_id = d.dept_id
237  INNER JOIN projects p
238  ON d.dept_id = p.dept_id
239  WHERE d.location IN ('Bangalore', 'Mumbai');
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	emp_name	location	proj_name
▶	Bhavna	Bangalore	Mobile App
	Bhavna	Bangalore	Cloud Migration
	Chirag	Bangalore	Mobile App
	Chirag	Bangalore	Cloud Migration
	Divya	Mumbai	Audit Automation
	Hina	Mumbai	Audit Automation

Question No. 27

Pair employees who work in the same city (avoid duplicate pairs and self-pairing).

```
241      -- 27
242
243 •   SELECT e1.emp_name AS employee_1,
244          e2.emp_name AS employee_2,
245          e1.city
246   FROM employees e1
247   INNER JOIN employees e2
248   ON e1.city = e2.city
249   AND e1.emp_id < e2.emp_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	employee_1	employee_2	city
▶	Bhavna	Chirag	Bangalore
	Divya	Hina	Mumbai
	Esha	Farhan	Delhi

Question No. 28

Show every project along with employees of that department; keep projects with no employees.

```
251      -- 28
252
253 •   SELECT p.proj_name, p.dept_id, e.emp_name
254   FROM projects p
255   LEFT JOIN employees e
256   ON p.dept_id = e.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	proj_name	dept_id	emp_name
▶	Payroll Revamp	10	Amit
	Cloud Migration	20	Chirag
	Cloud Migration	20	Bhavna
	Mobile App	20	Chirag
	Mobile App	20	Bhavna
	Audit Automation	30	Hina
	Audit Automation	30	Divya
	Brand Refresh	40	Farhan
	Data Lake	70	NULL